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| <b>5<br/>Number<br/>theory</b> | 5.1 | <b>An understanding of the division theorem and its application to the Euclidean Algorithm and congruences.</b>                                                                       | <b>Students should be able to apply the algorithm to find the highest common factor of two numbers.</b>                                                                                                                                                                                                                                                                                                                                                                                |
|                                | 5.2 | <b>Bezout's identity.</b>                                                                                                                                                             | <b>Students should be able to use back substitution to identity the Bezout's identity for two numbers.</b>                                                                                                                                                                                                                                                                                                                                                                             |
|                                | 5.3 | <b>Modular arithmetic. Understanding what is meant by two integers <math>a</math> and <math>b</math> to be congruent modulo an integer <math>n</math>. Properties of congruences.</b> | <p><b>The notation <math>a \equiv b \pmod{n}</math> is expected.</b></p> <p><b>Knowledge of the following properties:</b></p> <p><math>a \equiv a \pmod{n}</math></p> <p>if <math>a \equiv b \pmod{n}</math> then <math>b \equiv a \pmod{n}</math></p> <p>if <math>a \equiv b \pmod{n}</math> and <math>b \equiv c \pmod{n}</math> then <math>a \equiv c \pmod{n}</math></p> <p><b>Addition and subtraction laws for congruences.</b></p> <p><b>Multiplication and power laws.</b></p> |
|                                | 5.4 | <b>Fermat's Little Theorem.</b>                                                                                                                                                       | <p>For example, students should be able to find the least positive residue of <math>4^{20}</math> modulo 7</p> <p>Proof is not required.</p>                                                                                                                                                                                                                                                                                                                                           |

| Topics                                               | What students need to learn: |                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
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|                                                      | Content                      |                                                                  | Guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>5</b><br><b>Number theory</b><br><i>continued</i> | 5.5                          | <b>Divisibility Tests.</b>                                       | <b>For divisibility by 2, 3, 4, 5, 6, 9, 10 and 11.</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|                                                      | 5.6                          | Solution of congruence equations.                                | <p>Conditions under which solutions exist should be known.</p> <p>Use of Bezout's identity to find multiplicative inverses.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                                                      | 5.7                          | Combinatorics: counting problems, permutations and combinations. | <p>The multiplicative principle of counting. Set notation is expected; the number of subsets of a set should be known. Addition and subtraction principles.</p> <p>Students should, for example, be able to determine the number of positive integers less than 1000 containing the digit 3 (at least once).</p> <p>Understanding both <math>{}^nP_r</math> and <math>{}^nC_r</math> and when to use them. For example, to determine how many ways there are to select a team of 11 players from a squad of 21</p> <p>(a) regardless of playing positions and</p> <p>(b) if positions matter</p> |